

EDA Findings Report: Historical Stock Market Trends (1970-2010s)

[Link to the notebook](#)

Summary of Findings

Our exploratory data analysis of historical stock prices, primarily focusing on the `close` price across different decades, reveals several key trends and observations:

* **Increasing Data Volume:** There's a clear and significant increase in the number of stock records available from the 1970s to the 2010s. This could indicate a growth in the overall number of publicly traded companies, improved data collection over time, or a combination of both.

- * 1970s: ~164K records
- * 1980s: ~1.47M records
- * 1990s: ~3.43M records
- * 2000s: ~6.57M records
- * 2010s: ~9.34M records

* **General Upward Trend in Average Price:** The mean `close` price shows a consistent upward trajectory from the 1970s (\$12.49) through the 2000s (\$122.44), followed by a slight decrease in the 2010s (\$65.28). This suggests overall market growth and increasing stock valuations over several decades.

* **Increased Volatility:** The standard deviation of `close` prices dramatically increases from the 1970s (\$46.99) to the 2010s (\$1850.13). This indicates a significant rise in market volatility and larger price fluctuations in more recent decades, possibly reflecting a more dynamic and interconnected global economy.

* **Expanding Price Ranges:** The maximum `close` prices observed also show a substantial increase across decades, with the 2000s and 2010s reaching significantly higher peaks than previous decades. This points to the emergence of higher-priced stocks and a general expansion of market capitalization.

* **Visual Confirmation:** The 'Average Monthly Close Price' line plots visually reinforce these trends, showing relatively calmer periods in earlier decades with gradual growth, transitioning to more aggressive growth patterns and pronounced fluctuations in later decades. The 2000s, for instance, likely exhibited the aftermath of the dot-com bust and the lead-up to the 2008 financial crisis, characterized by sharp movements.

Interesting Trends, Seasonal Effects, and Anomalies

* **Long-Term Bull Market:** The most prominent trend is the overall growth of stock prices from the 1970s to the 2010s, indicative of a prolonged bull market across several economic cycles.

* **Decade-Specific Growth spurts:** Each decade seems to have its own growth characteristics. The 2000s, despite economic downturns, showed the highest average and maximum close prices, reflecting certain high-growth sectors or companies dominating the market.

* **Volatility as a Constant:** While increasing, volatility itself became a more consistent feature of the market in later decades, suggesting that large price swings are now a more regular part of stock market behavior.

- * **Lack of Obvious Intra-Year Seasonality:** At the aggregated level of average monthly close prices, there wasn't strong evidence of clear, recurring intra-year seasonal patterns influencing stock prices uniformly across all decades. Broader economic trends and events appear to be more dominant.
- * **Anomalies:** The outputs hint at significant market events (e.g., market corrections, bubbles, or crashes) being embedded within the data, leading to the large standard deviations and maximum price ranges, especially in the 2000s.

Initial Hypotheses on Factors Affecting Stock Prices

Based on this preliminary analysis, several initial hypotheses can be formulated regarding factors that might affect stock prices:

1. **Economic Growth and Recessions:** Periods of strong economic growth likely correlate with rising average stock prices and potentially lower volatility, while recessions or financial crises contribute to sharp declines and increased volatility (e.g., the potential impact of the 2008 crisis alluded to in the 2000s analysis).
2. **Technological Advancement and Innovation:** The significant increase in maximum prices and overall market value in later decades might be driven by rapid technological advancements, leading to the emergence of highly valued companies and industries (e.g., the tech boom).
3. **Monetary Policy and Interest Rates:** Low-interest rate environments could encourage investment in stocks, driving up prices, while higher rates might make fixed-income investments more attractive, potentially dampening stock market enthusiasm.
4. **Global Events and Geopolitical Stability:** Major global events (e.g., wars, pandemics, political instability) could introduce significant market volatility and impact investor confidence, leading to sharp price movements.
5. **Market Sentiment and Investor Behavior:** Beyond fundamentals, collective investor sentiment, herd behavior, and speculative activities might contribute to the observed increases in volatility and extreme price ranges.
6. **Regulatory Changes:** Shifts in financial regulations or industry-specific policies could impact company valuations and investor confidence, thereby affecting stock prices.

This analysis highlights the dynamic nature of the stock market and the interplay of various economic and social factors over time. A more granular analysis, including `open`, `high`, `low`, and `volume` data, would provide richer insights into these hypotheses.